
StarNet: Image Segmentation of Astronomy Images with the Segment Anything Model (SAM)

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Abstract

This work explores the application of the Segment Anything Model (SAM) to deep-field images from the CANDELS survey obtained with the Hubble Space Telescope. SAM was selected for its adaptability to unseen datasets and its demonstrated success in other fields such as medical imaging. We evaluate the performance of multiple pre-trained SAM variants and inference strategies, using Intersection over Union (IoU) as the main evaluation metric. We also develop an image pre-processing pipeline that converts the astronomical data into standard image formats. Despite the limited dataset, SAM shows promising results, with inference on split image patches significantly outperforming full-image inference and detecting a greater number of objects. However, the robustness of SAM in astronomical applications requires further investigation. Including high-resolution data from the James Webb Space Telescope (JWST) could enhance model generalisation ahead of upcoming deep-field surveys such as the Legacy Survey of Space and Time (LSST).

1. Introduction

The field of astronomy has undergone a massive data revolution in recent years. Space-based observatories such as the Hubble and James Webb Space Telescopes have revolutionised our view of the Universe, detecting even the faintest of stars and galaxies at frequencies that are inaccessible from Earth. This, however, comes at a cost. The development of better technology means higher spatial resolution and thus more objects per unit area within each image. This is particularly true for deep field images, which

can contain hundreds of thousands to millions of galaxies. As such, individual image processing and labelling by astronomers is extremely time-consuming as well as possibly subject to human biases. Automated methods are needed to process this large volume of data and save time for astronomers, particularly with upcoming sky surveys such as the Legacy Survey of Space and Time (LSST) with expected data volumes of ~ 500 PB (Ivezić et al., 2019).

Computer vision is an active area of study in machine learning, focusing on image analysis tasks such as object classification and image segmentation. To date, deep learning has been successfully applied to such tasks in several fields, like medical imaging, for example. Applications to astronomical images are still in their earliest stages, but have already proven to be incredibly useful. The first deep learning models in astronomy were developed based on the classic convolutional neural networks (CNNs; Dieleman et al., 2015) for image-level classification tasks. More recently, improvements in deep learning have allowed image segmentation and object detection to be possible, with novel architectures such as U-Net and Mask Region-based CNNs (Mask R-CNNs; Burke et al., 2019). Despite these developments, these models are still limited by the requirement of a large amount of labelled data, which in many cases is difficult and time-consuming to obtain. Computer vision models developed for astronomy-specific tasks should provide a robust and efficient method to perform these tasks in a time-efficient manner, allowing astronomers to concentrate on detailed analysis of individual objects once determined.

In this paper, we propose the adaptation of the recent Segment Anything Model (SAM) to the segmentation of deep-field galaxy images from the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey (CANDELS) obtained with the Hubble Space Telescope. Our contributions include a benchmark comparison of pre-trained SAM variants and the creation of carefully annotated ground-truth masks from the survey images, enabling effective fine-tuning and evaluation of segmentation performance. The paper is divided as follows: Section 2 reviews key advances in computer vision applications in astronomy over the past decade; Section 3 describes the Segment Anything Model (SAM) and the fine-tuning methods employed; Section 4

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describes the CANDELS galaxy images used, along with the preprocessing pipeline and data augmentation methods applied to create the full dataset for training; Section 5 outlines the SAM variants tested and the evaluation metrics; Section 6 presents the main quantitative and qualitative results; Section 7 discusses the strengths and limitations of each model; and finally, Section 8 summarizes the main findings and outlines directions for future work.

2. Computer Vision in Astronomy

The adoption of deep learning in astronomy began with image-level classification, driven by the success of convolutional neural networks (CNNs) across a range of computer vision tasks. A key milestone was the Galaxy Challenge, an international competition based on the Galaxy Zoo dataset, where participants were tasked with building the best models to predict galaxy morphologies from annotated images. A winning entry by [Dieleman et al. \(2015\)](#) introduced a CNN architecture tailored to this task, employing data augmentation through rotated image copies to enable the model to learn rotation-invariant features. Their model reproduced human classifications with near-perfect accuracy ($> 99\%$). Building on this, [Dai & Tong \(2018\)](#) applied a variant of residual networks (ResNets) to the second Galaxy Zoo dataset, where deeper residual blocks enabled the model to capture faint and complex structures, achieving improved performance over the [Dieleman et al. \(2015\)](#) model. These advances demonstrate the potential of deep neural networks to automate morphological classification while significantly reducing the manual workload for astronomers.

However, these models are limited to assigning only one classification per image. Advances in observational facilities, including increased sensitivity to fainter structures, have resulted in astronomical images commonly featuring crowded fields containing hundreds to thousands of objects. This shifted the focus toward image segmentation and object detection techniques, which first isolate individual objects within an image. Additional classification can be applied to each object, thereby effectively addressing the complexity of dense astronomical images.

A notable example is the Morpheus algorithm developed by [Hausen & Robertson \(2020\)](#), based on the U-Net architecture originally introduced by [Ronneberger et al. \(2015\)](#) for biomedical image segmentation. Morpheus performs semantic segmentation, in which each pixel in an image is assigned a class label, allowing it to simultaneously detect and classify objects in deep field images from instruments such as the Hubble Space Telescope. It was specifically designed to operate directly on astronomical data in their native Flexible Image Transport System (FITS) format, enabling multi-band processing across a range of wavelengths. This marked a major advance in astronomical computer vi-

sion by facilitating automated, pixel-level analysis directly on raw data, preserving image quality and capturing faint details at scale. However, even these models cannot distinguish overlapping objects of the same type. Instance segmentation models such as Mask Region-based CNNs (Mask R-CNNs; [He et al., 2017](#)) address this by detecting, classifying, and generating masks for individual objects in images. As such, [Burke et al. \(2019\)](#) adapted the Mask R-CNN model specifically for astronomical applications, demonstrating improved detection and masking of highly blended sources in crowded survey fields.

Despite significant advances in computer vision for astronomy, existing models still rely heavily on large, carefully annotated datasets for training. The increasing complexity and volume of astronomical data highlights the need for models capable of learning representations from fewer labelled examples and adapting efficiently to new tasks. This motivates the use of state-of-the-art computer vision models from computer science for astronomical applications, which we present in this work.

3. Methods

3.1. Segment Anything Model (SAM)

In this work, we use the base Segment Anything Model (SAM) as the foundation for our segmentation algorithm on astronomical images. SAM is a cutting-edge, general-purpose segmentation model developed by Meta AI ([Kirillov et al., 2023](#)), designed to identify and mask a wide variety of objects in images without prior task-specific training. Inspired by natural language processing (NLP) models,

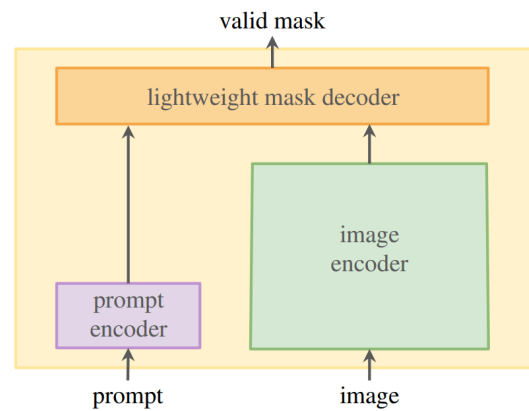


Figure 1. Illustration of the Segment Anything Model (SAM) architecture ([Kirillov et al., 2023](#)). The image is processed by the image encoder, while a user-provided prompt is passed to the prompt encoder. The resulting embeddings are combined in the mask decoder to generate segmentation masks for objects detected in the image.

SAM is promptable. It can generate segmentation masks based on user-provided inputs such as points, boxes, or reference masks. This approach parallels the prompt-response structure of language models, where generalisation across tasks is achieved through next-token prediction. SAM was trained on an exceptionally large dataset comprising approximately 1.1 billion masks across 11 million images, giving it strong generalisation capabilities (Kirillov et al., 2023). As a result, it can detect and segment numerous objects within a single image simultaneously.

Its architecture is composed of three main components: an image encoder, a prompt encoder, and a mask decoder. Together, these allow the model to operate flexibly across tasks and image domains, producing high-quality masks with minimal task-specific fine-tuning. The image encoder, a Vision Transformer (ViT), processes the images into embeddings that preserve spatial structure. The prompt encoder translates user inputs (e.g. points, boxes, or masks) into embeddings that guide the segmentation. Sparse prompts, such as points and boxes, are encoded as positional embeddings, while dense prompts are processed using convolutional layers. This allows SAM to flexibly respond to various prompt types and even generate multiple valid masks from a single prompt, enabling it to distinguish overlapping or nested objects. The mask decoder combines the image and prompt embeddings with a learned output token to produce segmentation masks. The model outputs a set of binary masks, each covering the full area of an individual object in the image. An illustration of the architecture of SAM is provided in Figure 1.

Due to its strong generalisation capabilities and ability to produce high-quality masks with minimal input, SAM has been widely applied in medical imaging. Its flexibility in segmenting complex and diverse features makes it effective for identifying anatomical structures such as tumours, necrotic tissue, and bacterial formations in prenatal scans (Cheng et al., 2023). Its promptable design allows clinicians to precisely and efficiently extract regions of interest. For similar reasons, SAM is well suited for application to astronomical images where annotated data is scarce and difficult to obtain (see Section 2). Additionally, astronomical images often contain hundreds to thousands of overlapping objects, requiring a model that can efficiently segment multiple instances within a single image. Given the anticipated volume of data from ongoing and future astronomical surveys (Ivezić et al., 2019), SAM presents an ideal solution for automated processing of this data as it requires minimal annotated data, thereby allowing astronomers to focus on deeper scientific analysis.

3.2. Model Finetuning

To improve performance on this task, we fine-tuned the models provided by MicroSAM. The training procedure follows the approach outlined in Archit et al. (2023), which is largely based on the codebase from Kirillov et al. (2023), with additional modifications including the integration of the `torch-em` library.

Training was performed on the processed and manually labelled images from the GOODS-S and GOODS-N CANDELS fields, as described in Section 4. The fine-tuning process began with segmentation masks generated by a pre-trained model, which were then corrected to produce accurate ground truth masks, following the annotation procedure detailed in Section 4. Although the available dataset was limited, comprising only 15 images divided into 10 training, 2 validation, and 3 test images, previous work by Archit et al. (2023) demonstrates that significant performance gains can be achieved with as few as 1 to 5 training images. This suggests that improvements are feasible even with such a small dataset. Nevertheless, data augmentation techniques were applied to expand the effective training set size by a factor of ten. More details on the finetuning are discussed in Section 5.

4. Data

The images used in this study are drawn from the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey (CANDELS), a large-scale Hubble Space Telescope (HST) program designed to probe galaxy formation and evolution across cosmic time. The CANDELS dataset is publicly available through the Mikulski Archive for Space Telescopes (MAST)¹. For this work, we selected images from two of the survey’s well-studied fields: the Great Observatories Origins Deep Survey North (GOODS-N) and South (GOODS-S). These fields provide deep, high-resolution observations across multiple wavelength bands, enabling the study of a diverse population of galaxies with varied morphological features.

These deep-field images were acquired using multiple wavelength filters spanning the optical to infrared regimes. To optimise wavelength coverage, we selected three filters and assigned each to a colour channel following the standard Hubble palette colour-mapping technique, which maps shorter wavelengths to blue, intermediate wavelengths to green, and longer wavelengths to red. The filters used in this work are F606W (optical, assigned to blue), F850LP (optical to near-infrared transition, assigned to green), and F160W

¹This work is based on observations taken by the CANDELS Multi-Cycle Treasury Program with the NASA/ESA HST, operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS5-26555.

(near-infrared, assigned to red). Example unprocessed images for each filter are provided in Appendix A.2.

In total, we downloaded 30 FITS frames covering four sectors in the GOODS-S and GOODS-N fields. In this section, we describe the preprocessing pipeline used to convert these frames into RGB images, the annotation procedure employed to generate inputs for the prompt encoder, and the data augmentation techniques applied to expand the available dataset.

4.1. Image Processing

As discussed in Section 2, astronomical imaging data are typically stored in the Flexible Image Transport System (FITS) format. However, most image segmentation models, including the Segment Anything Model (SAM), are designed to operate on standard image formats such as PNG or JPEG. Consequently, a preprocessing step was required to convert the FITS images into standard three-channel (RGB) representations, while preserving as much of the original information as possible. This preprocessing pipeline was implemented entirely in Python, utilising commonly used libraries such as `astropy`, `numpy`, and `scipy`.

The images used in this work were acquired using two instruments aboard the *Hubble Space Telescope* (HST): the Advanced Camera for Surveys (ACS), which provides a resolution of 30 milliarcseconds (mas) per pixel, and the Wide Field Camera (WFC), with a resolution of 60 mas per pixel. To ensure consistency across the dataset, the ACS images were downsampled by a factor of two to match the spatial resolution and dimensions of the WFC images. Subsequently, denoising techniques were applied to suppress background noise and enhance the visibility of faint structures. Subsequently, denoising techniques were applied to suppress background noise and enhance the visibility of faint structures. Initially, percentile clipping was performed to remove extreme pixel values and outliers commonly associated with detector noise, retaining only pixels within the central 95% of the intensity distribution. This step facilitated the clear highlighting of galaxies without removing significant structural information. To further refine the images, a median filter with a kernel size of 10 was applied, reducing residual noise and edges, thereby improving the clarity of galaxy features for subsequent segmentation. Following these steps, each image channel was individually normalised to the [0,1] range according to:

$$p_{norm} = \frac{p - \min(p)}{\max(p) - \min(p)}$$

where p denotes the intensity of a given pixel. This process was applied separately to each filter image. Finally, the normalised filter images were combined into a single RGB composite image by assigning each filter to a colour channel

as described before: F160W to red, F850LP to green, and F606W to blue. An example of a processed image is shown in Figure 2 for a section of the GOODS-S field. A total of 23 images were produced from this pre-processing pipeline.

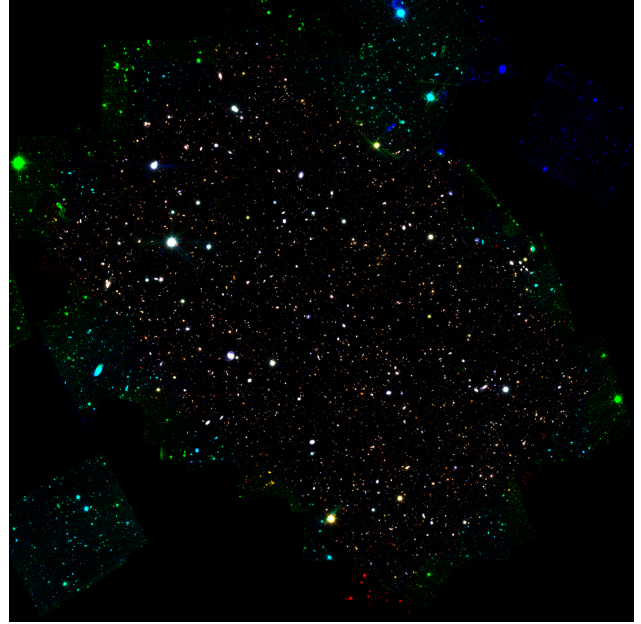


Figure 2. Sample image of Section 33 of the GOODS-S CANDELS field after pre-processing has been applied.

4.2. Image Annotation

Following preprocessing, each image was annotated with a ground truth mask. Initially, inference was performed on the images to generate preliminary segmentation masks. These masks were subsequently refined through manual correction, which involved removing false positives, adjusting object boundaries, and adding any missed objects.

This annotation procedure was applied to 15 images from the GOODS-N and GOODS-S fields, constituting nearly the entire dataset. To enhance detection accuracy, each image was divided into 16 equal-sized segments, and inference was conducted on each segment using the ViT-B model. This resulted in 240 segmentations subjected to manual review and correction. The refinement process required approximately one to two hours per image.

4.3. Data Augmentation

Given the limited size of the available dataset, we employed data augmentation techniques to improve the robustness of our model. Augmentation improves the model’s ability to recognise galaxies under variations in orientation, scale, and brightness. The augmentation methods applied include:

- Random rotations: either fixed 90° rotations or random

angles within $\pm 30^\circ$.

- Horizontal flips: applied with a 40% probability.
- Vertical flips: applied with a 20% probability.
- Small affine transformations: including scaling by $\pm 10\%$ and translations up to 5% of the image size.
- Brightness, contrast, and gamma adjustments: to simulate variations in exposure and lighting conditions.
- Coarse dropout: randomly masking small regions of the image to help the model handle incomplete data.
- Random cropping: selecting crops from predefined square and rectangular sizes ranging from 256 to 512 pixels.

We carefully tuned the parameters of these augmentations and evaluated how much they impact the segmentation results. Some examples of the generated images from our augmentation script are provided in Appendix A.3.

5. Experiments

Three types of image segmentation are compared on the images produced by the pre-processing pipeline described in the previous section. The first is a direct inference using the pre-trained models from Kirillov et al. (2023), including those finetuned for microscopy tasks as presented in Archit et al. (2023). The second inference method follows a multi-step approach. First, the image is split into smaller patches. Each patch is then upscaled, and inference is performed on the upscaled patches. Finally, the resulting masks are down-scaled and recombined into the final segmentation of the original image. This method also uses pre-trained models from (Kirillov et al., 2023) and (Archit et al., 2023). The final inference method employs a model finetuned specifically on the data presented in this work.

Training was conducted on a local machine equipped with an NVIDIA Titan X GPU. The ViT-B model was fine-tuned over 5 epochs, requiring approximately 2 to 3 hours of runtime. A second model was finetuned, the vit-b-lm model, which is specialised for light microscopy tasks.

The different segmentation methods were applied to the augmented dataset and compared to assess their performance. The metric used for comparison is the Intersection over Union (IoU), a standard metric in image segmentation. IoU is defined as the ratio of the number of pixels common to both the true and predicted masks to the number of pixels present in either mask, and is expressed mathematically as:

$$IoU = \frac{Mask_{true} \cap Mask_{pred}}{Mask_{true} \cup Mask_{pred}} \quad (1)$$

In addition to this numerical comparison, the segmentation results were also examined qualitatively to assess the relative strengths and weaknesses of each method.

6. Results

12 different models were tested on the dataset. The models tested include vit-b, vit-l, vit-b-lm (a fine-tuned vit-b for light microscopy data) and vit-l-lm. Each model was assessed using both direct inference and inference on split images. Additionally, direct inference performance was compared between the vit-b and vit-b-lm models trained on the original dataset versus those trained on the augmented dataset. The comparison of the IoUs of a subset of these models is shown in Figure 3. The direct inference using vit-b-lm, vit-l and vit-l-lm is omitted here, along with inference on split images using vit-l-lm. A complete graph is available in Appendix A.1.

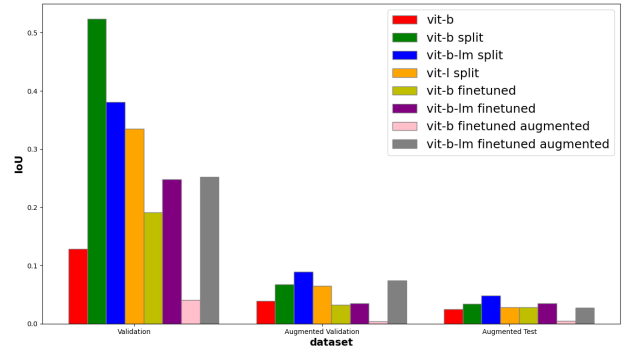


Figure 3. Bar graph comparing the performances of some of the models on the validation set, the augmented validation set and the augmented test set. The best-performing model on the validation set is the vit-b using split images, followed by the vit-b-lm on split images. For the augmented datasets, the vit-b-lm on split images performs best. Its performance is almost matched by the vit-b-lm fine-tuned with an augmented dataset. Overall, the worst-performing model is the vit-b finetuned with augmented data.

The best-performing model on the base validation set is by far the vit-b using split images, reaching an IoU of over 0.5. However, the performance of nearly all models declines significantly on the augmented data, with the vit-b model among the most affected. On the augmented data, the performances of the vit-b-lm using split images and the vit-b-lm finetuned using augmented data become more competitive. Furthermore, the vit-b model fine-tuned on the augmented data exhibits the worst performance by far. It is also worth noting that across all parts of the dataset, the vit-b model fine-tuned on the original dataset consistently performs equal to or better than the base vit-b model using direct inference. In addition, the vit-b-lm model fine-tuned with augmented data performs comparably or better than the vit-b model when trained solely on the original dataset.

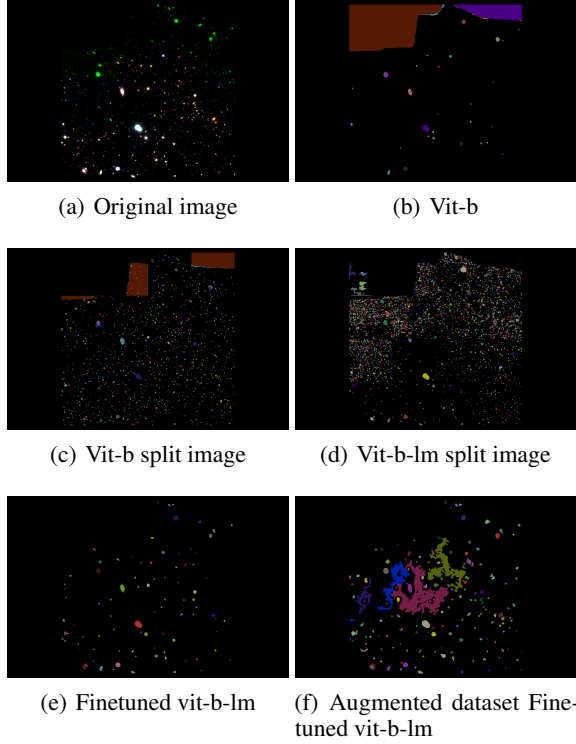


Figure 4. Output masks produced by different models when performing inference on sector 14 of the GOODS-N survey. The direct inference detects fewer objects but captures their shapes accurately. In contrast, the other models identify significantly more objects, with the split-image inference detecting the greatest number of elements.

A qualitative comparison is presented in Figure 4, showing inference results on an image of sector 14 from the GOODS-N survey. The figure compares the masks generated by vit-b using direct inference with those produced by vit-b and vit-b-lm using split-image inference, including both fine-tuned vit-b-lm models. A more comprehensive comparison is provided in Appendix A.1.

It is clear that the vit-b model detects by far the fewest objects, closely followed by the fine-tuned vit-b-lm model. However, the vit-b model, like the vit-b on split images and the vit-b-lm finetuned on augmented data, appears to exhibit a false positive problem with large empty areas. In contrast, both split-image inference models result in the highest number of detections, consistent with the high mean IoU performance on the non-augmented data observed in earlier experiments.

7. Discussion

The results presented here suggest that inference on split images is the greatest source of performance improvement, although it doesn’t appear to be very robust. Such a large

improvement likely arises from the model’s difficulty in detecting very small objects, which constitute the majority of targets in this dataset. This is confirmed by the qualitative results, where split inference models detect significantly more elements than other methods. Thus, zooming in on image regions appears to strongly enhance segmentation performance.

It is encouraging that the finetuning does appear to improve performance in most cases. This is likely due to the fine-tuned models detecting more objects while also suffering less from false positives. However, further experiments with larger datasets are needed to determine whether greater improvements are possible, possibly identifying a model that balances the strengths of split-image inference and fine-tuned approaches.

One aspect that remains difficult to explain is the poor performance of the vit-b model finetuned with augmented data. This is particularly intriguing given that both the vit-b model fine-tuned on the base dataset and the vit-b-lm model finetuned on the same augmented data perform reasonably well. A possible explanation is that the model was unable to adapt to the augmented data, whereas the vit-b-lm model, with some training for similar tasks (microscopy image segmentation), was more capable of adapting quickly. This would mean this model could simply need a longer training time, but this remains to be confirmed through further experimentation.

Worse performance on the augmented dataset was to be expected, but the extent of this decline was far greater than expected. One possible cause is the relatively low quality of the masks used for training. However, this can only be confirmed by comparing the performances of each model with different types of data augmentation and high-quality masks during training.

8. Conclusion

In this work, we applied various pre-trained variants of the Segment Anything Model (SAM) to segment deep-field galaxy images from the CANDELS survey. Our experiments indicate that the most significant performance improvements come from performing inference on split image patches. This increases object detection at the cost of increased false positives. On the other hand, direct inference models detect fewer objects, resulting in overall poorer segmentation performance. Despite the limited size of our dataset, fine-tuned SAM models showed promising improvements, particularly in identifying smaller objects. However, the poor performance of the vit-b model fine-tuned with augmented data highlights the need for careful evaluation of augmentation methods.

Further research is needed to assess the robustness of these

models. Expanding the dataset to include more diverse and higher-resolution images would improve generalisation. Once publicly available, deep field survey images from the James Webb Space Telescope (JWST) could be used for this purpose. In the meantime, semi-supervised learning techniques offer a promising approach for training on limited datasets. Additionally, improving annotation quality and automating prompt generation would reduce manual effort and enhance training reliability.

A particularly time-intensive component was the image pre-processing pipeline required to convert the raw FITS images into RGB format. Adapting the SAM model to work directly with FITS images, potentially using VAE-based encoders, could provide better efficiency and accuracy. Adding a galaxy morphology classification layer would also allow comparisons with existing models like Morpheus (Hausen & Robertson, 2020), extending the use of SAM for astronomical applications.

Finally, future image segmentation models, including updated versions of SAM (e.g. SAM 2; Ravi et al., 2024) or alternative architectures like Mask R-CNNs (Richards et al., 2024), may further enhance performance on astronomical imaging tasks and should be explored in future work.

9. Code Availability

All the data and programs used for the experiments presented here are available on GitHub at <https://github.com/Nael3004/StarNet>. There is no dedicated documentation but comments and a readme are available to help users. Most of the code used to produce the results are also available there.

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A. Appendix

A.1. Extended Model Comparison

More models were experimented with both quantitatively and qualitatively. The image of sector 14 from the test set was used to test twelve models, vit-b, vit-b-lm, vit-l, vit-l-lm, each using both direct inference and split images. Additionally, versions of vit-b and vit-b-lm finetuned using the base dataset and an augmented dataset were tested as well. The quantitative comparison can be found in 5 and the qualitative study can be seen in 6.

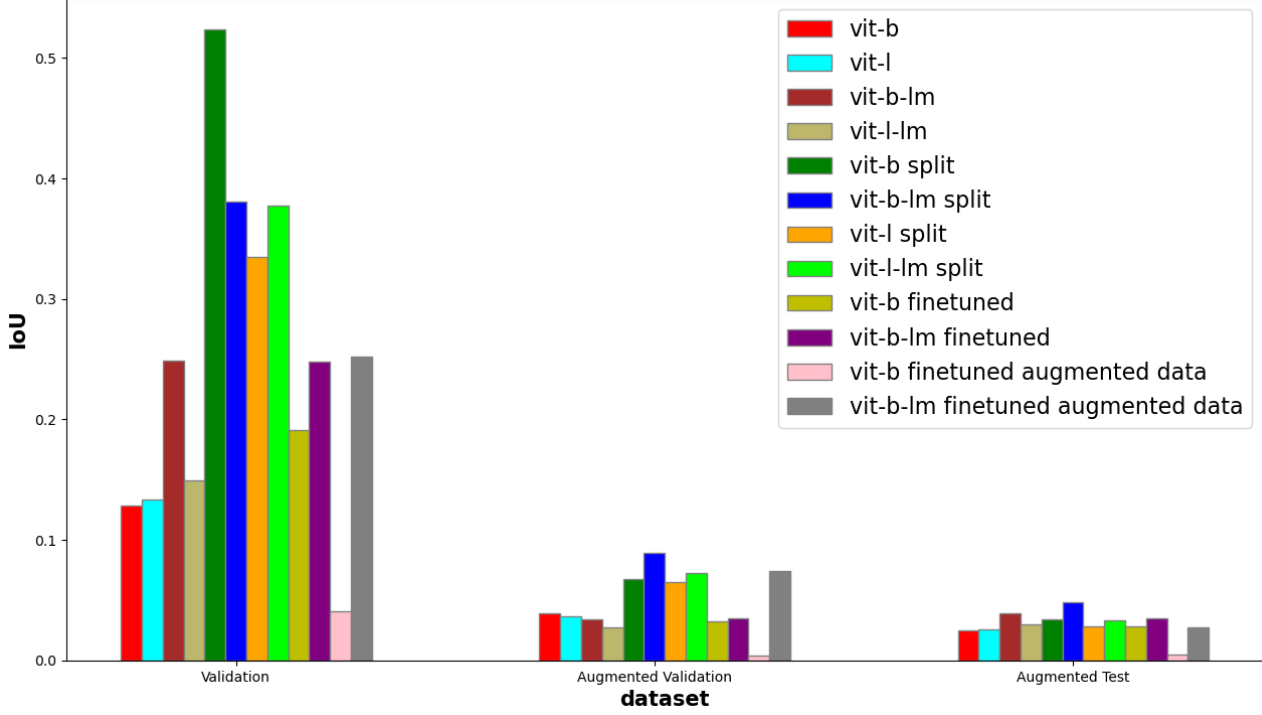


Figure 5. Bar graph comparing the performances of the models on the validation set, the augmented validation set and the augmented test set. On the Validation set the best performing model is clearly the vit-b using split images followed by the vit-b-lm and vit-l-lm. On the augmented validation and test sets however, all models’ performances worsen greatly which leaves the vit-b-lm on split images as the most performant. Overall the worst performing model is the vit-b finetuned with augmented data.

As can be seen in figure 5, the trend observed when comparing the vit-b model and both finetuned models based on it seems to hold for vit-b-lm as its finetuned versions perform about as well or better. It is intriguing that vit-b-lm performs much better than vit-l-lm as a bigger model is assumed to perform better. But model size being uncorrelated to performance seems to be a trend here both for the direct inference and the inference on split images.

From the masks displayed in Figure 6, it can be seen again that model size doesn’t appear to influence performance very strongly, since the results given by models only differ in size, like vit-b and vit-l look very similar. It can also be seen that the augmented dataset finetuned vit-b detects nothing or almost nothing in the image. That would explain its extremely poor performance, although it does not help to understand the cause behind this.

It is also interesting to observe that of 5 models that have false positives over large empty areas, 4 are the basic models not trained for microscopy. This suggests that one of the biggest gains of the finetuning performed in Archit et al., 2023 is the model becoming able to recognise such areas as empty. This also makes this mistake which reappears in the vit-b-lm finetuned with augmented data even more intriguing.

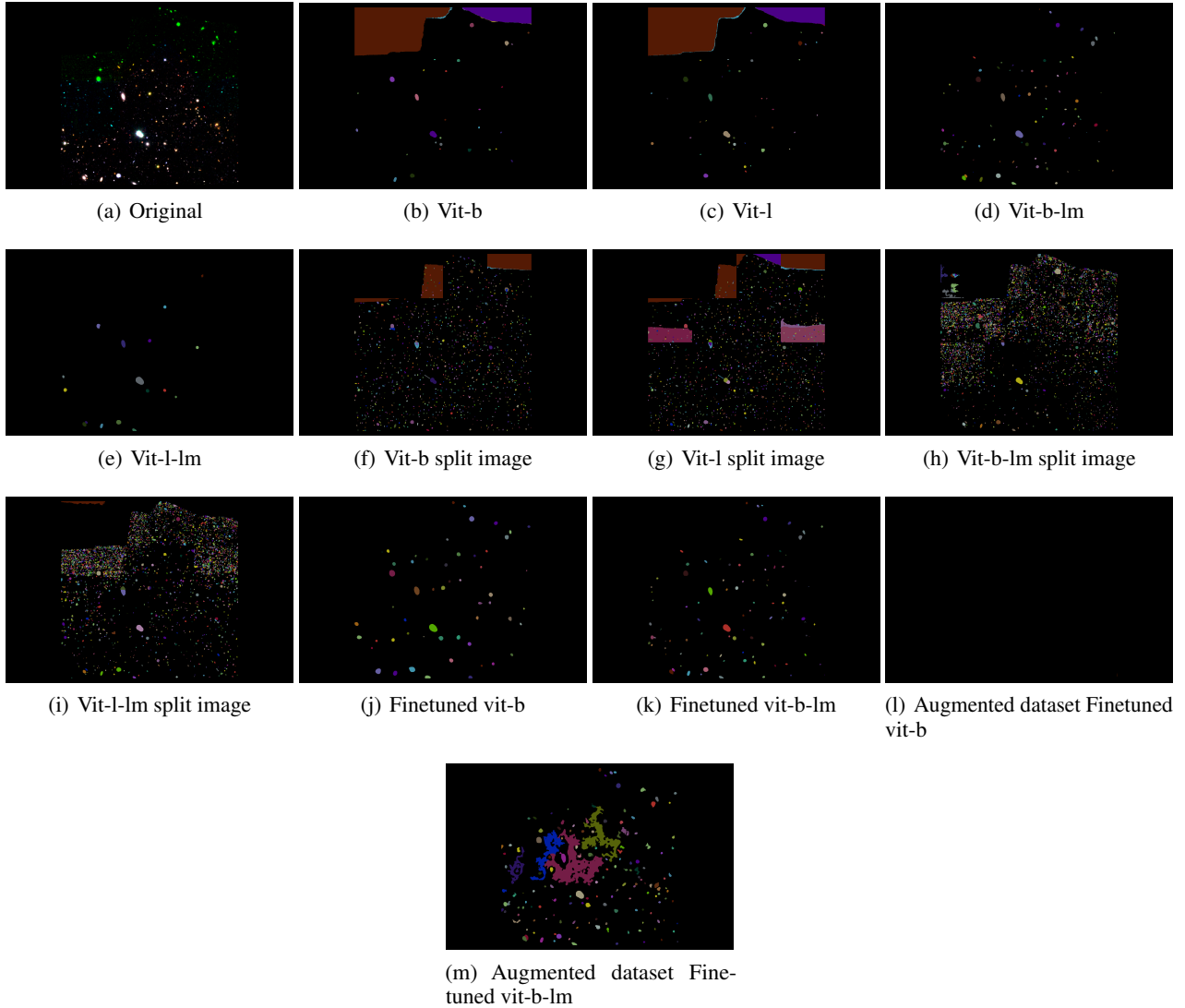


Figure 6. Outputted mask of each model when doing inference on sector 14 of the GOODS-N survey. The vit-b finetuned using augmented data detects almost nothing. The other models that detect the fewest objects are all the direct inference models save for vit-b-lm. On the opposite hand, all split image models appear to detect many objects, the most of all models here.

A.2. Examples of Individual Filter Images

The three filters were selected to provide broad wavelength coverage while adhering to the widely used ‘‘Hubble palette’’, in which shorter wavelengths are mapped to blue and longer wavelengths to red. The F606W filter lies in the optical regime and is sensitive to stellar continuum emission, offering strong contrast for identifying star-forming regions. The F850LP filter is a long-pass filter that spans the transition from the optical to the near-infrared; assigning it to the green channel enhances the visibility of older stellar populations and dust features that are less prominent at shorter wavelengths. The F160W filter, situated in the near-infrared, captures the reddest light in the dataset and is particularly effective for detecting high-redshift galaxies, whose emission has been redshifted to longer wavelengths. This filter combination enables the construction of colour-composite images that are both perceptually intuitive and scientifically representative of the underlying astrophysical structures.

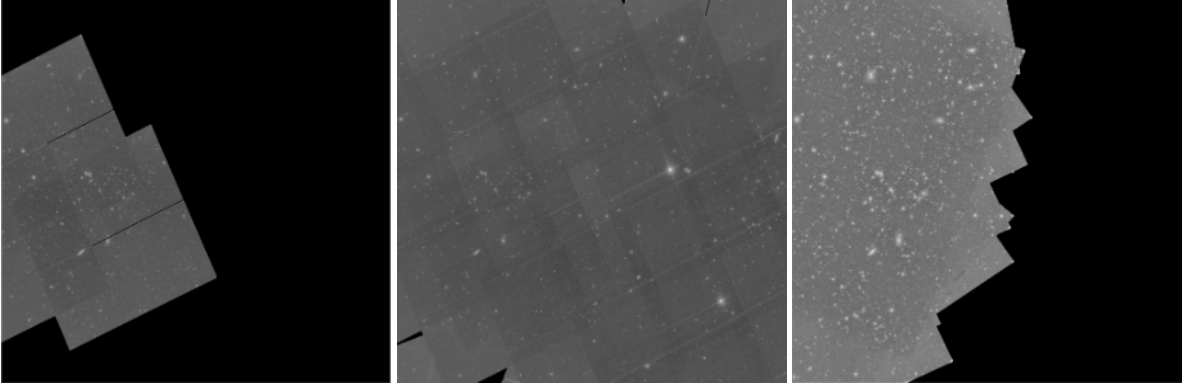


Figure 7. From left to right: F606W (optical, ‘‘blue’’), F160W (infrared, ‘‘red’’) filter images for the same field.

A.3. Examples of Augmented Images

This section illustrates sample augmentations applied to the pre-processed images, as detailed in Section 4.3. Figure 8 displays some sample augmentations applied to these images, which extend the amount of data that can be applied to the SAM models for training and testing.

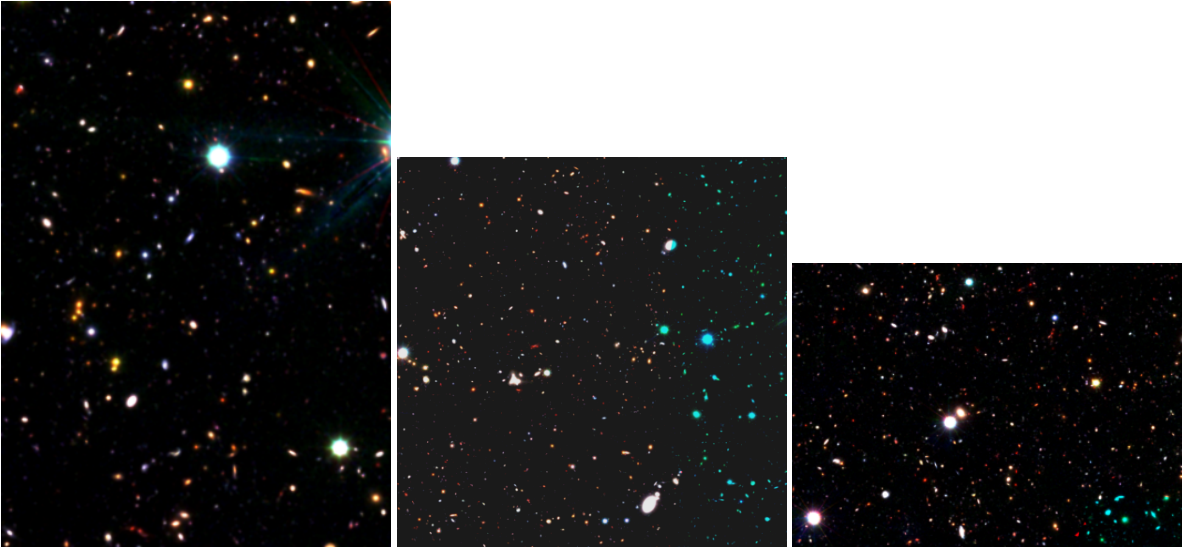


Figure 8. Examples of augmented image tiles generated by data augmentation.